Fogging System



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **Medium** 4 labs

Covers these lab items: Fogging and blackout; Misting

What it is used for

A fogging or misting system disperses fine droplets to provide controlled humidity, cooling, or an approved treatment application. Correct setup and nozzle control help distribute mist evenly while reducing exposure, runoff, and equipment blockage.

Main parts

- Water or approved solution reservoir
- Pump and pressure regulator
- Filter and supply tubing
- Fogging or misting nozzles
- Control valves, timer, and pressure gauge if fitted
- Mounting brackets and electrical controls where applicable

Before you start

1. Read the system instructions and identify the approved liquid and intended application.
2. Inspect the reservoir, pump, tubing, filters, nozzles, and electrical connections for damage or leakage.
3. Confirm that the water or treatment solution is suitable for the system and intended site.
4. Clean the filter and check that nozzles are clear and correctly installed.
5. Verify that the pressure setting and droplet size are suitable for the application and surrounding conditions.
6. Wear the personal protective equipment required for the liquid and ensure people, animals, and sensitive equipment are protected.

How to use

1. Fill the reservoir with clean water or the approved solution according to the instructions.
2. Secure the reservoir lid and check that the supply lines are connected correctly.
3. Open the required valves and set the recommended operating pressure.
4. Start the pump or control system and inspect each nozzle for an even mist pattern.
5. Adjust nozzle direction and flow to cover the intended area without excessive wetting or drift.
6. Set the timer or application duration according to the approved operating plan.
7. Monitor pressure, liquid level, leakage, wind or airflow conditions, and the uniformity of mist distribution.
8. Stop the system if a nozzle blocks, a line leaks, or mist drifts into an unintended area.
9. Switch off and isolate power before cleaning nozzles or servicing the pump.

10. Drain, clean, and store the system according to the manufacturer's instructions and the liquid used.

⚠ Safety

- Use only liquids approved for the system and intended application; never improvise chemical mixtures.
- Follow the safety data sheet for any treatment solution and use the specified protective equipment.
- Avoid inhaling mist and keep people or animals away when the application requires exclusion.
- Depressurize the lines before loosening fittings or removing nozzles.
- Disconnect electrical power before maintenance and keep electrical components protected from moisture.
- Do not operate fogging equipment near ignition sources when using flammable liquids.

Care & storage

- Clean reservoir, filters, and nozzles at the recommended intervals to prevent blockage and contamination.
- Inspect tubing, seals, pump connections, and electrical protection for wear or leaks.
- Flush the system after using a treatment solution as instructed by the manufacturer.
- Store the unit clean and dry, and protect nozzles from dust and physical damage.

Common mistakes to avoid

- Excessive pressure can create unwanted drift or a mist pattern unsuitable for the application.
- Using dirty water or an unfiltered solution can block fine nozzles.
- Applying mist in unsuitable wind or airflow can carry droplets beyond the target area.
- Failing to flush the system after treatment use can cause corrosion, residue, or cross-contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.